MAKE THINGS HAPPY !

4 Digital Inputs (ISO1211) + 4 SPST Relays - IoTextra Octal4

Industrial Grade DI & SPST Relay Module



4 Digital Inputs (ISO1211) + 4 SPST Relays - IoTextra Octal4 (hereinafter **IoTextra Octal4** (Rev. 3.02) is an eight-channel digital I/O module, containing four digital inputs and four relays.
Digital inputs use the Texas Instruments ISO1211D isolated digital input receiver — an integrated alternative to a classic optocoupler front-end, requiring no field-side power supply.
Input voltage range is **field-configurable per channel** via solder jumpers, selectable among 4 standard industrial classes: 12 / 18 / 24 / 36 VDC
Module has four Panasonic APAN3105, or Hongfa HF49FD relays.
Contact form — **SPST (1 Form A)**.
All digital input channels have individual galvanic isolation with an isolation rating of 2500 Vrms (capacitive isolation barrier).

The module provides indication of the digital input and relay channel status.

Up-to-date technical information about the module is available on GitHub.

SOFTWARE POWER:

The **IoTextra Octal4** module offers native support for popular IoT platforms:

Node-RED Home Assistant TASMOTA Blynk IoTflow

This hardware compatibility unlocks the full potential of the IoTflow platform. It enables you to:

- Create custom **Node-RED** nodes effortlessly.
- Integrate via **MQTT** with third-party global platforms like **Blynk**.
- Scale your project from a single node to a complex industrial network.

Integrate with **Tasmota** for reliable **MQTT**-based telemetry and remote control — support for **IoTextra** modules, including **IoTextra Octal4**, is described here.

There are two modes for using the **IoTextra Octal4** module:

- ***GPIO Mode***. In this mode, reading the state of the input signals and controlling the output signals are performed using the **AP0-AP7** signals on the **HOST-P** connector.
- ***I^2C Mode***. In this mode, reading the state of input signals and controlling output signals are performed via the **I^2C** bus using the I/O expander installed on the module (TCA9534 or a compatible chip). Up to 16 modules can be connected to one **I^2C** bus.

Combined Use: It is also possible to use the **AP0-AP7** signals for some channels through the **HOST-P** connector and the I/O expander for the remaining channels.





This module is designed for systems requiring a mix of isolated field-signal monitoring and relay-based load switching, with a per-channel adjustable input voltage range.

APPLICATIONS:

🏭 **Industrial Automation — Sensor & Actuator I/O**
Digital inputs read field sensors, limit switches, and proximity switches at standard industrial levels — 12VDC, 24VDC, and other configurable classes per channel; relays switch contactors, indicator lamps, or small actuators.
Recommended software: Node-RED, IoTflow

⚡ **IIoT — Machine Status Monitoring**
Inputs monitor e-stop loops, door interlocks, and voltage-presence signals across mixed voltage classes on the same module; relays drive local alarms or feed isolated status signals to a PLC.
Recommended software: Tasmota

🌱 **Smart Farming — Irrigation & Climate Control**
Float switches and limit switches connect to the digital inputs; relays activate pumps, valves, or heaters.
Recommended software: Node-RED, IoTflow

🏠 **Smart Home — Entrance & Equipment Control**
Inputs read gate/door position sensors and buttons; relays operate electric strikes, lighting, or general-purpose loads.
Recommended software: Home Assistant

FEATURES:

- Compatibility with major microcontrollers
- Module power supply is 5VDC, which feeds a 3V3 regulator for powering the logic section
- Includes a 3V3 Indicator LED
- Protection against reverse power supply polarity
- 4 independent, galvanically isolated digital input channels (TI **ISO1211D** isolated digital input receiver), no field-side power supply required
- Isolation rating: 2500 Vrms
- **Input voltage range is field-configurable per channel** via solder jumpers, selectable among 4 standard classes: 12 / 18 / 24 / 36 VDC — see INPUT RANGE CONFIGURATION below
- Polarity-independent (bipolar) digital input
- 4 SPST (1 Form A) type relays (**Panasonic APAN3105** / **Hongfa HF49FD**)
- Switching Voltage: 250VAC / 30VDC
- Max. switching current: 5A
- Coil resistance [±10%] (at 20°C/68°F): 227Ω
- Insulation resistance: 1000MΩ (at 500VDC)
- Breakdown voltage: between open contacts — 1000Vrms for 1min; between contact and coil — 3000Vrms for 1min
- Surge breakdown voltage (Initial) (between contacts and coil): 6000V
- Standard electromechanical relay (non-latching): logical "1" (2.4-5V) — relay off (NC); logical "0" (0-0.9V) — relay activation (NO)
- The module features LED indicators for the status of digital input channels and relay state
- Reading the input signal states and controlling the relays is done using AP0-AP7 through the HOST-P12 connector and/or via the I²C bus through the I/O expander
- The AP4 line can alternatively carry the INT signal from the I/O expander via jumpers SB5/SB6 (by default AP4 controls relay K1, interrupt not used)
- The I/O expander used in the module is the **TCA9534** or a compatible chip
- The expander's **I²C** address (**A2-A0**) is set using DIP microswitch:

TCA9534	0	1	0	0	A2	A1	A0	x
TCA9534A	0	1	1	1	A2	A1	A0	x

- Connection to the module via the **I²C** bus is done through **Qwiic©** connectors or through pins 11 (**SDA**) and 12 (**SCL**) of the **HOST-P12** connector.
- Transient suppression and electrostatic discharge protection of signals on **Qwiic©** connectors is done using a TVS diode assembly (ESD protection)
- Module size is 47x56 mm. The module has mounting holes allowing it to be installed in the base module or in a **Raspberry Pi**

INPUT RANGE CONFIGURATION:

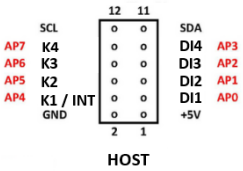
Each digital input channel has three solder jumpers (JT) forming three parallel branches between the SENSE node and the field terminal: one direct short (no resistor), one through RT1 (2.7kΩ), and one through RT3 (3.6kΩ). Closing exactly one jumper selects that branch as the active RTHR:

Voltage Range	Active branch	RTHR	VIH	VIL	Margin (VIH to nominal)
12V DC	short (no resistor)	0Ω	8.20V	7.00V	+3.80V
18V DC	RT1 ‖ RT3 branch*	1.543kΩ	12.96V	11.71V	+5.04V
24V DC	RT1 branch	2.7kΩ	17.55V	15.88V	+6.45V
36V DC	RT3 branch	3.6kΩ	21.42V	19.58V	+14.58V

** 18V is the only range where both RT1 and RT3 branches are closed simultaneously (parallel combination).*
RS = 6.8kΩ always installed, no jumper.

HOST CONNECTOR:

The pinouts for these **HOST** connectors are shown in the image below:



HOST

The **HOST** connector is used differently depending on how the microcontroller interacts with the **IoTextra Octal4** module:

1) ***Standalone Use:*** In this mode, the input signal states and control of the output signals are available via the **HOST** connector and the **I²C** bus using the I/O expander.
2) ***Smart Use:*** In this mode, an **IoTsmart** microcontroller module is inserted into the **HOST** connector. Reading the input signal states, control of the output signals and the 5V module power supply are done through the **HOST** connector. Power is supplied from the **IoTsmart** module. Below is a photo of the **IoTextra Octal4** module with the **IoTsmart** module:


IoTextra Octal4 with **IoTsmart ESP32-S3**

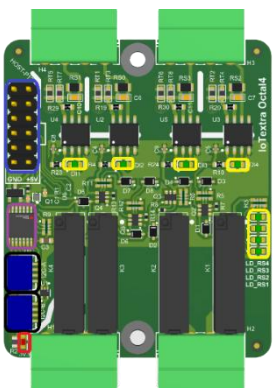

IoTextra Octal4 with **IoTsmart RP2350A**


IoTextra Octal4 with **IoTsmart RP2040**

3) ***Mezzanine Use:*** In this mode, the **IoTextra Octal4** module is installed in the base module. The state of the input signals and the output signal control are available through the **HOST** connector.

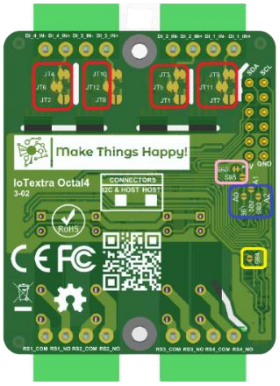
COMPONENT LAYOUT:

The image below shows the component and marking layout on the **top-side** of the **IoTextra Octal4** module:



It is possible to obtain input signal status and control output signals via the **HOST** connector (highlighted in blue).
The **Qwiic©** connectors are highlighted in black.
The I/O expander is highlighted in purple.
There is a 3V3 power indicator. The power indicator are highlighted in red in the **top-side** image.
The I/O signal indicators are shown in yellow in the figure.

JUMPERS:



The image below shows the component and marking layout on the **bottom-side** of the **IoTextra Octal4** module:

Each channel's three jumpers correspond to: short (direct, no resistor), RT1 (2.7kΩ branch), and RT3 (3.6kΩ branch) — always close exactly one, except for 18V where both RT1 and RT3 are closed together:

Range	Short	RT1	RT3
12V	**CLOSED**	open	open
18V	open	**CLOSED**	**CLOSED**
24V	open	**CLOSED**	open
36V	open	open	**CLOSED**

Short = direct jumper (no resistor, 0Ω). RT1 = 2.7kΩ branch. RT3 = 3.6kΩ branch. 18V is the only range where two jumpers (RT1 and RT3) are closed together.

Per-channel jumper numbers:

DI1 — Short=JT7, RT1=JT9, RT3=JT11
DI2 — Short=JT1, RT1=JT3, RT3=JT5
DI3 — Short=JT8, RT1=JT10, RT3=JT12
DI4 — Short=JT2, RT1=JT4, RT3=JT6

SB1-SB3 (highlighted in blue) set the I²C address bits (A0-A2) of the I/O expander.

SB5/SB6 (shown in pink) set the function of AP4 (SB6 closed = AP4 controls relay K1, default; SB5 closed = AP4 is INT from the I/O expander).

SB4 (highlighted in yellow) is a jumper that allows powering devices connected through the Qwiic© connector from +3V3EXT. By default, the jumper is absent.

CONFIGURATION TABLES:

The bottom-side of the module provides confiration information about **I²C** connectors:



EXTERNAL SIGNAL CONNECTION:

Removable terminal blocks with a 3.5 mm pitch are recommended for connecting external input signals to the **H3-H4** connectors on the module. The pin layout is marked on the **bottom-side** of the module.

The relays are connected to **H1- H2** connector.

ACCESSORIES:

The following accessories may be required for using the module:

- A set of four removable terminal blocks with a 3.5mm pitch for terminal blocks **H1-H4**
- A set of two standoffs and four screws for mounting the module into the **IoTbase** series base module.
- Cable for the **Qwiic©** connector